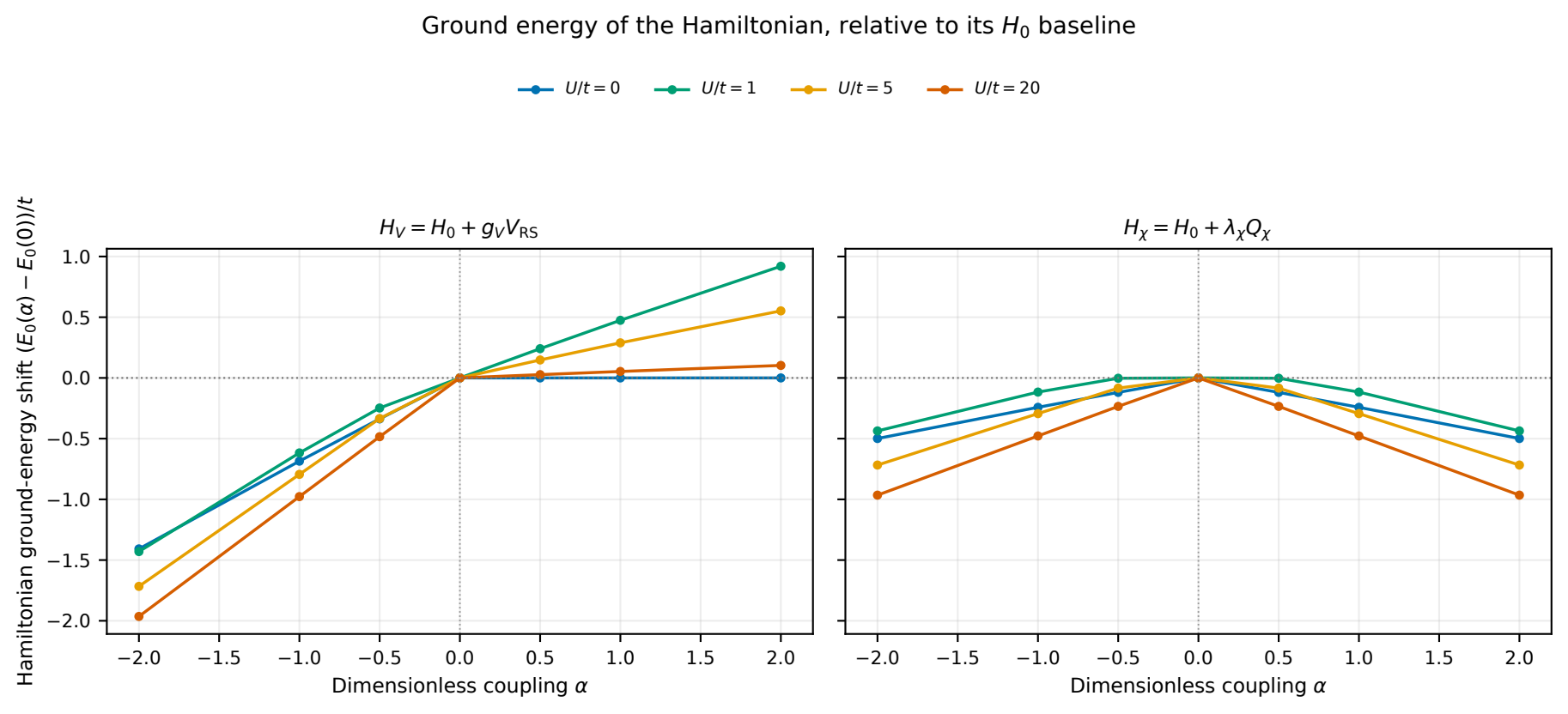
Ground energy of the Hamiltonian, relative to its H_0 baseline
$U/t = 0$
$U/t = 1$
$U/t = 5$
$U/t = 20$
$H_V = H_0 + g_V V_{\mathrm{RS}}$
$H_\chi = H_0 + \lambda_\chi Q_\chi$
Hamiltonian ground-energy shift $(E_0(\alpha) - E_0(0))/t$
Dimensionless coupling α
Dimensionless coupling α
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0